Water Pollution

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Water pollution is the contamination of water sources by substances which make the water unusable for drinking, cooking, cleaning, swimming, and other activities. Pollutants include chemicals, trash, bacteria, and parasites. All forms of pollution eventually make their way to water. Water bodies include lakes, rivers, oceans, aquifers, reservoirs and groundwater. Water pollution results when contaminants are introduced into these water bodies. Water pollution can be attributed to one of four sources: sewage discharges, industrial activities, agricultural activities, and urban runoff including stormwater.

Sources of Water Pollution

Sources of water pollution are either point sources or non-point sources.

1) Point Sources

Point sources have one identifiable cause, such as a storm drain, a wastewater treatment plant or an oil spill. Because point sources are located at specific places, they are fairly easy to identify, monitor, and regulate.

Point Source of water pollution

2) Non-point Sources

Nonpoint sources are broad and diffuse areas, rather than points, from which pollutants enter bodies of surface water or air. Examples include runoff of chemicals and sediments from cropland, livestock feedlots, logged forests, urban streets, parking lots, lawns, and golf courses.

Nonpoint sediment pollution eroded from farmland flows into streams

Types of Water Pollution

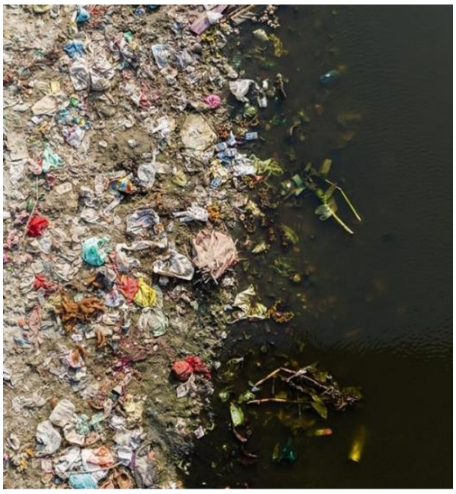
1) Ground-water Pollution

Groundwater gets polluted when contaminants—from pesticides and fertilizers to waste leached from landfills and septic systems—make their way into an aquifer, rendering it unsafe for human use. Ridding groundwater of contaminants can be difficult to impossible, as well as costly. Once polluted, an aquifer may be unusable for decades, or even thousands of years. Groundwater can also spread contamination far from the original polluting source as it seeps into streams, lakes, and oceans. Common sources of ground water pollution include septic tanks, industries like textile, chemical and tanneries, deep well injections, mining, etc.

2) Surface-water Pollution

Covering about 70 percent of the earth, surface water constitutes our oceans, lakes, rivers, streams, etc. Major sources of surface water pollution are:

1. Sewage: emptying drains and sewers
2. Industrial effluents: industrial waste containing toxic chemicals, acids, alkalis, salts and radioactive waste
3. Synthetic detergents: in washing and cleaning
4. Agrochemicals: fertilizers, insecticides and pesticides
5. Oil: spillage into sea during drilling and shipment
6. Waste heat: from industrial discharges increases water temperature and affects the distribution and survival of sensitive species



Surface water Pollution

Types of Contaminants

Water pollutants can be classified as organic pollutants, inorganic pollutants, pathogens, suspended solids, nutrients and agriculture pollutants, thermal, radioactive, and other pollutants. Organic and inorganic pollutants are mainly discharged from industrial effluents and sewage into the water bodies.

1) Organic Contaminants

The following are the types of organic contaminants that are responsible for water pollution

- Detergents
- Food processing waste: fats, grease, oxygen demanding substances
- Insecticides and herbicides: organohalides
- Petroleum hydrocarbons: fuels, lubricants and fuel combustion products
- Volatile organic compounds: industrial solvents
- Chlorinated solvents (PCBs, trichloroethylene)
- Drug pollution
- Personal hygiene and cosmetic products

2) Nitrogen and Phosphorus Compounds

Addition of compounds containing nitrogen and phosphorus helps in the growth of algae and other plants that consume DO after death. Foul smelling gases are produced under anaerobic conditions. Excess growth or decomposition of plant material changes CO2 concentrations, thereby affecting water pH. These changes in DO, oxygen and temperature change the physicochemical properties of water.

3) Inorganic Compounds

The following inorganic contaminants are responsible for water pollution. They are

- Acidity caused by industrial discharge (SO_2)
- Ammonia from food processing waste

- Chemical waste
- Fertilizers containing nutrients (nitrates and phosphates)
- Heavy metals from moto vehicles and acid mine drainage
- Silt/sediment

4) Pathogens

Wastewater sewage contain several pathogenic and non-pathogenic microorganisms and viruses that can cause water-borne diseases such as cholera, dysentery, typhoid, jaundice etc. Coliform bacteria do not cause an actual disease, but is used as a bacterial indicator of water pollution. High levels of pathogens may result from on-site sanitation systems (septic tanks, pit latrines) or inadequately treated sewage discharges. Combined sewers in certain cities discharge untreated sewage during rain storms that can result in contamination. Pathogen discharge can also be caused by poorly managed livestock operations.

5) Macroscopic Pollution

They are large, visible items polluting water, also called floatables or marine debris found in open seas, including

- Trash/garbage: discarded by people, or washed by rainfall into storm drains and eventually reaching surface waters
- Nurdles: small ubiquitous waterborne plastic pellets
- Shipwrecks: large, derelict ships

6) Thermal Pollution

Thermal pollution, sometimes called "thermal enrichment", is the degradation of water quality by any process that changes ambient water temperature. A common cause of thermal pollution is the use of water as a coolant by power plants and industrial manufacturers. Thermal pollution can also be caused by the release of very cold water from the base of reservoirs into warmer rivers. Fish and other organisms adapted to particular temperature range can be killed by an abrupt change in water temperature (either a rapid increase or decrease) known as "thermal shock".

7) Radioactive Substances

Radioactive waste is any pollution that emits radiation beyond what is naturally released by the environment. It’s generated by uranium mining, nuclear power plants, and the production and testing of military weapons, as well as by universities and hospitals that use radioactive materials for research and medicine. Radioactive waste can persist in the environment for thousands of years, making disposal a major challenge.

Table 20-1 Major Water Pollutants and Their Sources

Type/*Effects*	Examples	Major Sources
Infectious agents (pathogens) *Cause diseases*	Bacteria, viruses, protozoa, parasites	Human and animal wastes
Oxygen-demanding wastes *Deplete dissolved oxygen needed by aquatic species*	Biodegradable animal wastes and plant debris	Sewage, animal feedlots, food-processing facilities, paper mills
Plant nutrients *Cause excessive growth of algae and other species*	Nitrates (NO_3^-) and phosphates (PO_4^{3-})	Sewage, animal wastes, inorganic fertilizers
Organic chemicals *Add toxins to aquatic systems*	Oil, gasoline, plastics, pesticides, fertilizers, cleaning solvents	Industry, farms, households, mining sites, runoff from streets and parking lots
Inorganic chemicals *Add toxins to aquatic systems*	Acids, bases, salts, metal compounds	Industry, households, mining sites, runoff from streets and parking lots
Sediments *Disrupt photosynthesis, food webs, other processes*	Soil, silt	Land erosion from farms and construction and mining sites
Heavy metals *Cause cancer, disrupt immune and endocrine systems*	Lead, mercury, arsenic	Unlined landfills, household chemicals, mining refuse, industrial discharges
Thermal *Make some species vulnerable to disease*	Heat	Electric power and industrial plants

Effects of Water Pollution

- Increase of oxygen demand: Demand of O_2 increases with addition of biodegradable organic matter, expressed as biological oxygen demand (BOD)
- Diseases: In humans, drinking or consuming polluted water in any way has many disastrous effects on our health. It causes typhoid, cholera, hepatitis and various other diseases.
- Destruction of Ecosystems: Ecosystems are extremely dynamic and respond to even small changes in the environment. Water pollution can cause an entire ecosystem to collapse if left unchecked.
- Biomagnification: Non-biodegradable waste biomagnifies, causing toxic effects at various levels of the food chain. Several chemicals such as DDT are not water soluble, and tend to accumulate in body lipids, building up at successive levels of the food chain.
- Eutrophication: Chemicals in a water body, encourage the growth of algae. These algae form a layer on top of the pond or lake. Bacteria feed on these algae and this decreases the amount of oxygen in the water body, severely affecting the aquatic life there.
- Effects the food chain: Disruption in food chains happens when toxins and pollutants in the water are consumed by aquatic animals (fish, shellfish etc) which are then consumed by humans.

Control of Water Pollution

The best way to protect streams from pollution is to prevent it at the source. The following are some of the methods that can be implemented to control water pollution.

- Judicious use of pesticides and fertilizers
- Use of nitrogen-fixing plants
- Prevent manure run-off into surface water, instead divert them into basins for settlement that can be used later as fertilizer
- Separate drainage of sewage and rain water to prevent overflow and contamination
- Planting trees would reduce pollution by preventing runoff

- Treatment of wastewater is essential to prevent pollution from point sources
- Parameters considered for water quality: BOD, chemical oxygen demand (COD), nitrates, phosphates, oil and grease, toxic metals
- Waste water should be treated properly by primary and secondary methods to reduce BOD, COD levels up to permissible levels for discharge